Lab LT 4.1 · Soap, salt and cold: what pulls DNA out of a strawberry?

Name _____ Period _____ Group # _____

1. Predict. Which group will score lowest? Why?

_____

2. Our tube. Score it with the rubric, then type all three into the class spreadsheet.

Mass _____ g	Rating (0–4) ____	Spooled? Y / N

0	**Nothing**	clear liquid; nothing on the toothpick
1	**Haze**	faint cloudy haze; nothing catches
2	**Wisps**	thin strands catch, then slide off
3	**Clump**	white layer; a clump lifts out
4	**Rope**	thick layer; a long rope lifts out, more left behind

3. Results. Copy from the Results tab of the spreadsheet.

#	Group	This class rating	All classes average	Real difference from control?
1	**Control** · standard mix			
2	**No soap** · 0% soap			
3	**Low soap** · 2% soap			
4	**No salt** · 0% salt			
5	**High salt** · 6% salt			
6	**Room-temp alcohol** · alcohol not iced			

4. Claim and evidence. Which variable changed the amount of DNA the most: soap, salt or alcohol temperature? Use two averages.

_____

_____

_____

5. Why trust the all-classes average more than your own class's rating?

_____

_____

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1 Group 1: Control

Unit 4 · Lesson 1 · LT 4.1 · Strawberry DNA

Your conditions: 10% soap · 1.5% salt · ice-cold alcohol

1. Weigh **10 g** of thawed strawberry (9–11 g) in a weigh boat. Write the mass.
2. Put it in the zip bag. Add **10 mL** from bottle **A · Standard**. Squeeze out the air and seal.
3. Mash for **1 minute**. Do not break the bag.
4. Pour through the coffee filter in the funnel until the tube holds about **2 cm**.
5. Raise your hand. Your teacher pours **ice-cold alcohol** slowly down the side of the tube. Do not shake it.
6. Wait **1 minute**. Score the tube with the rubric, then do the spool test: turn a toothpick 5 times at the boundary and lift it out.
7. Type your mass, rating and Y/N in the spreadsheet: your period column, row 1.

0	**Nothing**	clear liquid; nothing on the toothpick
1	**Haze**	faint cloudy haze; nothing catches
2	**Wisps**	thin strands catch, then slide off
3	**Clump**	white layer; a clump lifts out
4	**Rope**	thick layer; a long rope lifts out, more left behind

Safety. Goggles on. Only your teacher pours the alcohol; it is flammable. Nothing at this station is eaten. Bag and filter in the trash; rinse the tube.

2 Group 2: No soap

Unit 4 · Lesson 1 · LT 4.1 · Strawberry DNA

Your conditions: 0% soap · 1.5% salt · ice-cold alcohol

1. Weigh **10 g** of thawed strawberry (9–11 g) in a weigh boat. Write the mass.
2. Put it in the zip bag. Add **10 mL** from bottle **B · No soap**. Squeeze out the air and seal.
3. Mash for **1 minute**. Do not break the bag.
4. Pour through the coffee filter in the funnel until the tube holds about **2 cm**.
5. Raise your hand. Your teacher pours **ice-cold alcohol** slowly down the side of the tube. Do not shake it.
6. Wait **1 minute**. Score the tube with the rubric, then do the spool test: turn a toothpick 5 times at the boundary and lift it out.
7. Type your mass, rating and Y/N in the spreadsheet: your period column, row 2.

0	**Nothing**	clear liquid; nothing on the toothpick
1	**Haze**	faint cloudy haze; nothing catches
2	**Wisps**	thin strands catch, then slide off
3	**Clump**	white layer; a clump lifts out
4	**Rope**	thick layer; a long rope lifts out, more left behind

Safety. Goggles on. Only your teacher pours the alcohol; it is flammable. Nothing at this station is eaten. Bag and filter in the trash; rinse the tube.

3 Group 3: Low soap

Unit 4 · Lesson 1 · LT 4.1 · Strawberry DNA

Your conditions: 2% soap · 1.5% salt · ice-cold alcohol

1. Weigh **10 g** of thawed strawberry (9–11 g) in a weigh boat. Write the mass.
2. Put it in the zip bag. Add **10 mL** from bottle **C · Low soap**. Squeeze out the air and seal.
3. Mash for **1 minute**. Do not break the bag.
4. Pour through the coffee filter in the funnel until the tube holds about **2 cm**.
5. Raise your hand. Your teacher pours **ice-cold alcohol** slowly down the side of the tube. Do not shake it.
6. Wait **1 minute**. Score the tube with the rubric, then do the spool test: turn a toothpick 5 times at the boundary and lift it out.
7. Type your mass, rating and Y/N in the spreadsheet: your period column, row 3.

0	**Nothing**	clear liquid; nothing on the toothpick
1	**Haze**	faint cloudy haze; nothing catches
2	**Wisps**	thin strands catch, then slide off
3	**Clump**	white layer; a clump lifts out
4	**Rope**	thick layer; a long rope lifts out, more left behind

Safety. Goggles on. Only your teacher pours the alcohol; it is flammable. Nothing at this station is eaten. Bag and filter in the trash; rinse the tube.

4 Group 4: No salt

Unit 4 · Lesson 1 · LT 4.1 · Strawberry DNA

Your conditions: 10% soap · 0% salt · ice-cold alcohol

1. Weigh **10 g** of thawed strawberry (9–11 g) in a weigh boat. Write the mass.
2. Put it in the zip bag. Add **10 mL** from bottle **D · No salt**. Squeeze out the air and seal.
3. Mash for **1 minute**. Do not break the bag.
4. Pour through the coffee filter in the funnel until the tube holds about **2 cm**.
5. Raise your hand. Your teacher pours **ice-cold alcohol** slowly down the side of the tube. Do not shake it.
6. Wait **1 minute**. Score the tube with the rubric, then do the spool test: turn a toothpick 5 times at the boundary and lift it out.
7. Type your mass, rating and Y/N in the spreadsheet: your period column, row 4.

0	**Nothing**	clear liquid; nothing on the toothpick
1	**Haze**	faint cloudy haze; nothing catches
2	**Wisps**	thin strands catch, then slide off
3	**Clump**	white layer; a clump lifts out
4	**Rope**	thick layer; a long rope lifts out, more left behind

Safety. Goggles on. Only your teacher pours the alcohol; it is flammable. Nothing at this station is eaten. Bag and filter in the trash; rinse the tube.

5 Group 5: High salt

Unit 4 · Lesson 1 · LT 4.1 · Strawberry DNA

Your conditions: 10% soap · 6% salt · ice-cold alcohol

1. Weigh **10 g** of thawed strawberry (9–11 g) in a weigh boat. Write the mass.
2. Put it in the zip bag. Add **10 mL** from bottle **E · High salt**. Squeeze out the air and seal.
3. Mash for **1 minute**. Do not break the bag.
4. Pour through the coffee filter in the funnel until the tube holds about **2 cm**.
5. Raise your hand. Your teacher pours **ice-cold alcohol** slowly down the side of the tube. Do not shake it.
6. Wait **1 minute**. Score the tube with the rubric, then do the spool test: turn a toothpick 5 times at the boundary and lift it out.
7. Type your mass, rating and Y/N in the spreadsheet: your period column, row 5.

0	**Nothing**	clear liquid; nothing on the toothpick
1	**Haze**	faint cloudy haze; nothing catches
2	**Wisps**	thin strands catch, then slide off
3	**Clump**	white layer; a clump lifts out
4	**Rope**	thick layer; a long rope lifts out, more left behind

Safety. Goggles on. Only your teacher pours the alcohol; it is flammable. Nothing at this station is eaten. Bag and filter in the trash; rinse the tube.

6 Group 6: Room-temp alcohol

Unit 4 · Lesson 1 · LT 4.1 · Strawberry DNA

Your conditions: 10% soap · 1.5% salt · room-temperature alcohol

1. Weigh **10 g** of thawed strawberry (9–11 g) in a weigh boat. Write the mass.
2. Put it in the zip bag. Add **10 mL** from bottle **A · Standard**. Squeeze out the air and seal.
3. Mash for **1 minute**. Do not break the bag.
4. Pour through the coffee filter in the funnel until the tube holds about **2 cm**.
5. Raise your hand. Your teacher pours **room-temperature alcohol** slowly down the side of the tube. Do not shake it.
6. Wait **1 minute**. Score the tube with the rubric, then do the spool test: turn a toothpick 5 times at the boundary and lift it out.
7. Type your mass, rating and Y/N in the spreadsheet: your period column, row 6.

0	**Nothing**	clear liquid; nothing on the toothpick
1	**Haze**	faint cloudy haze; nothing catches
2	**Wisps**	thin strands catch, then slide off
3	**Clump**	white layer; a clump lifts out
4	**Rope**	thick layer; a long rope lifts out, more left behind

Safety. Goggles on. Only your teacher pours the alcohol; it is flammable. Nothing at this station is eaten. Bag and filter in the trash; rinse the tube.